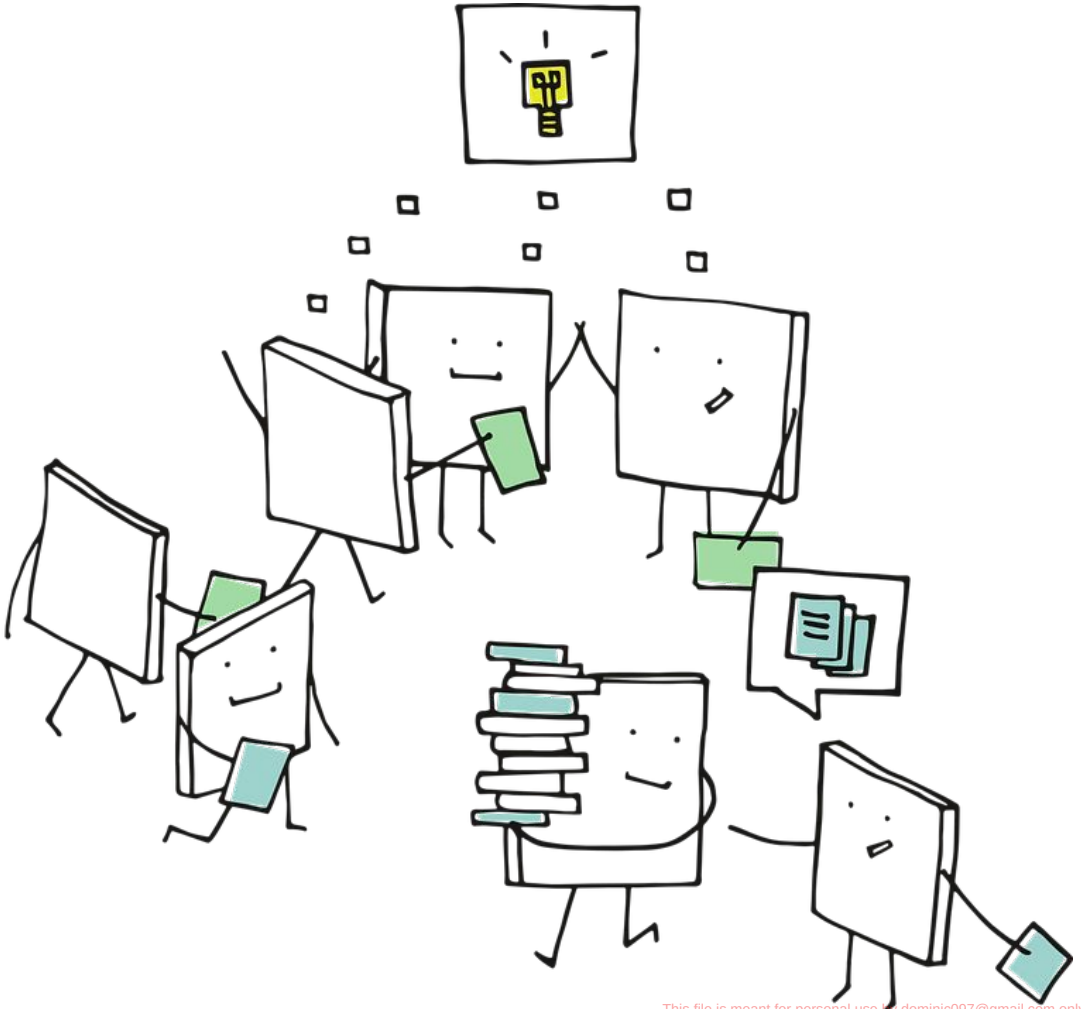

Statistical NLP

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What we will learn

What is NLP?

Tokenization & Vectorization

Text Normalization

Working with Web data

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What we will learn

Parts of Speech (PoS) tagging

Dependency Parsing

Named Entity Recognition (NER)

Word2Vec Embeddings

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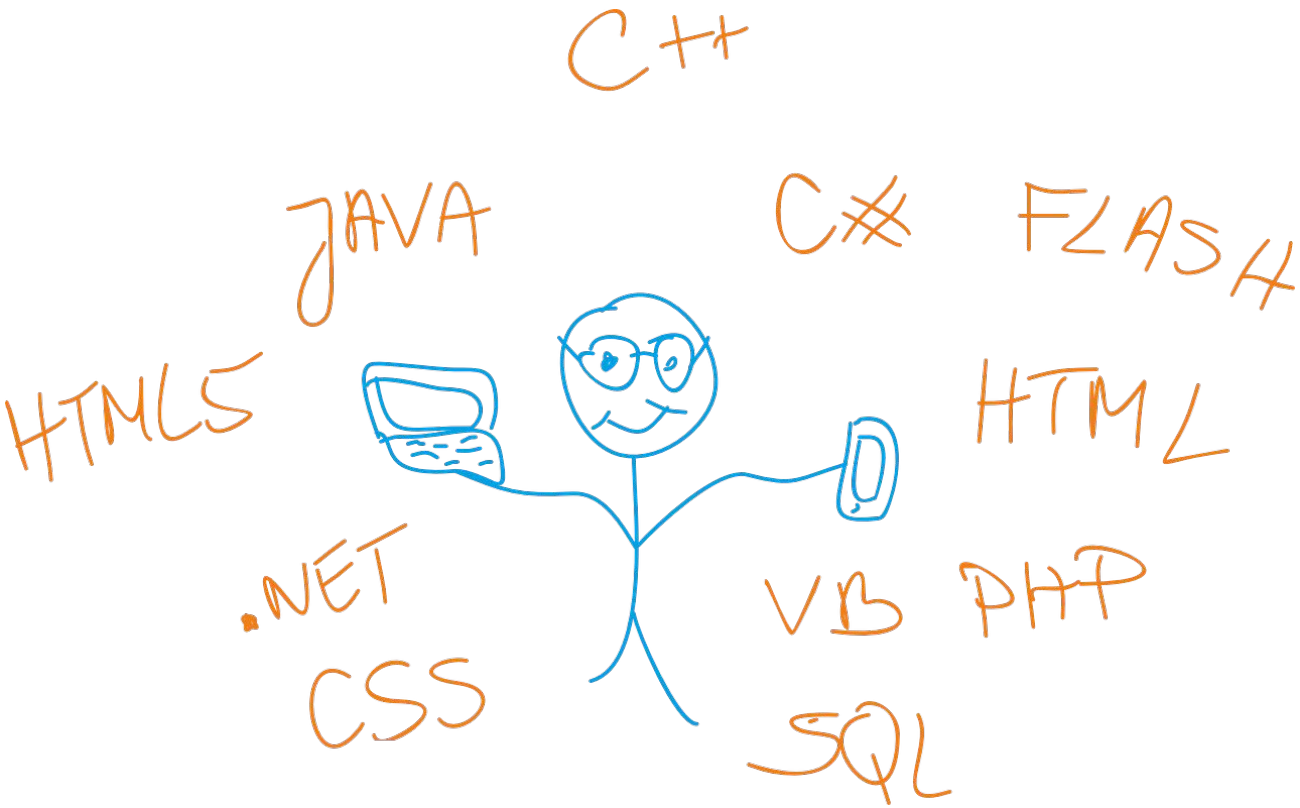

Computers have their own languages!

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And we humans have our own languages ...called Natural Language

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If we have to ask questions to Computer . . like

- What is the next train to Delhi after 9pm?
- What genes are associated with Diabetes?
- What are people liking about our new product?
- How much time I have before my next meeting?

We have to convert it to Computer's language

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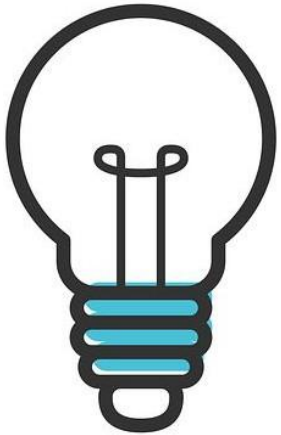

Wouldn’t it be nice if computers can also use natural language . . .

That’s the goal of NLP

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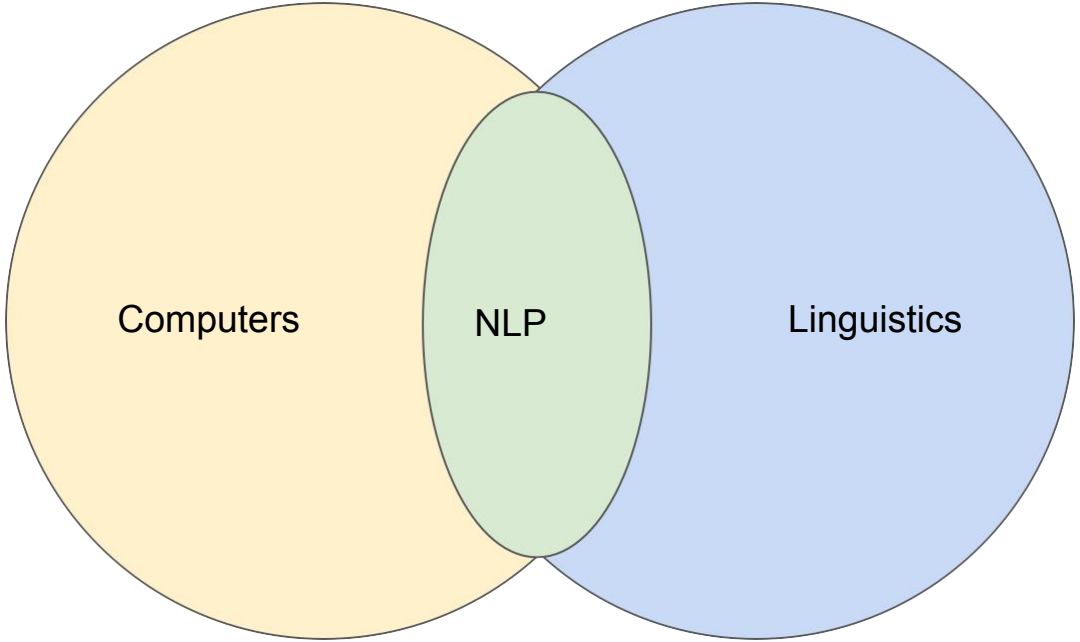

NLP is a multidisciplinary field involving

Computer Science and Linguistics

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Why is NLP hard?

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Red Tape holds up New Bridges.

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The court to try shooting defendant.

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He approached many banks for the loan.

Delhi is situated on the banks of Yamuna river.

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So how do we build logic to do NLP?

- It’s very hard for us humans to build the rules . . (to code)
- Hence, we commonly use <u>Machine Learning</u> for NLP

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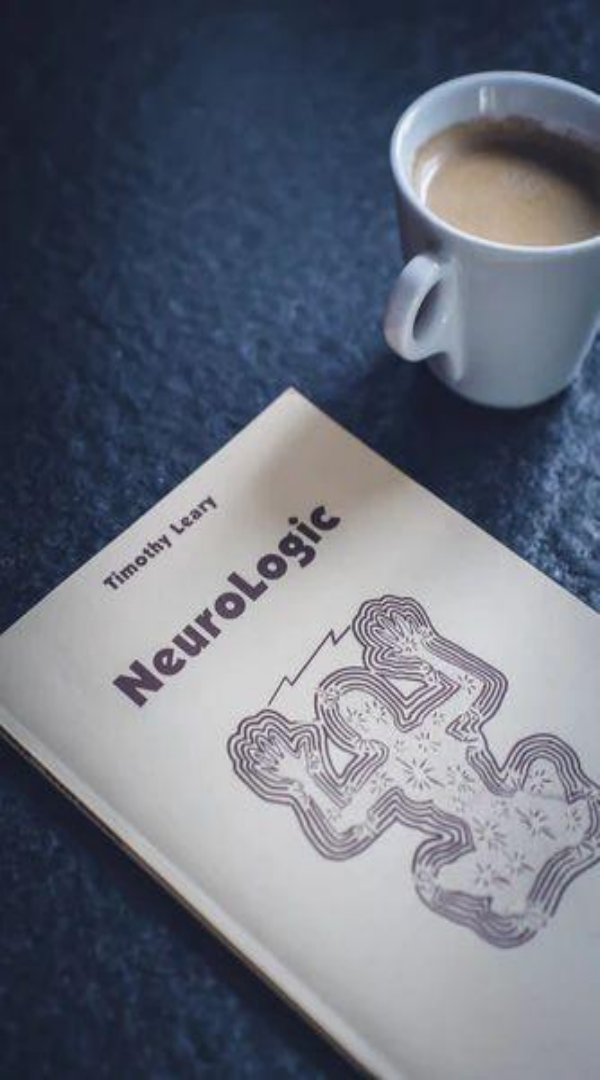

NLP presents a huge opportunity …

as most organization have humongous amount of textual data but struggling to get value out of it.

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Healthcare

Banking & Insurance

Fashion

Manufacturing

Retail

Travel

Social Media

NLP is useful for every industry …

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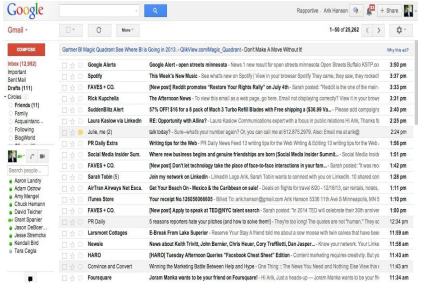
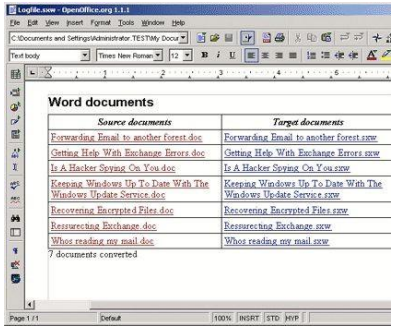


Emails

Social Media Updates

Chat interactions



Office documents

podcast

...other audio data

Some examples of Language Data

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Applications of NLP

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Google Home, Alexa etc

Digital assistants

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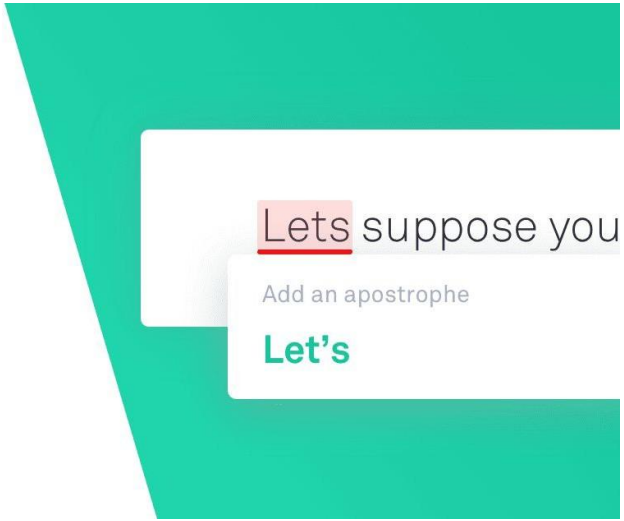
Help us write better

Improved communication

grammarly



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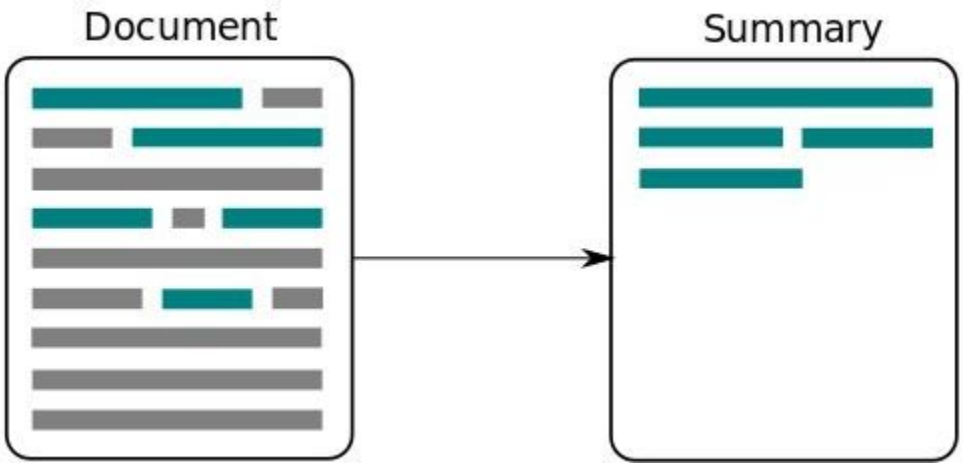

Text summarization

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Tendulkar playing cricket

Describe a picture

(Image Captioning)

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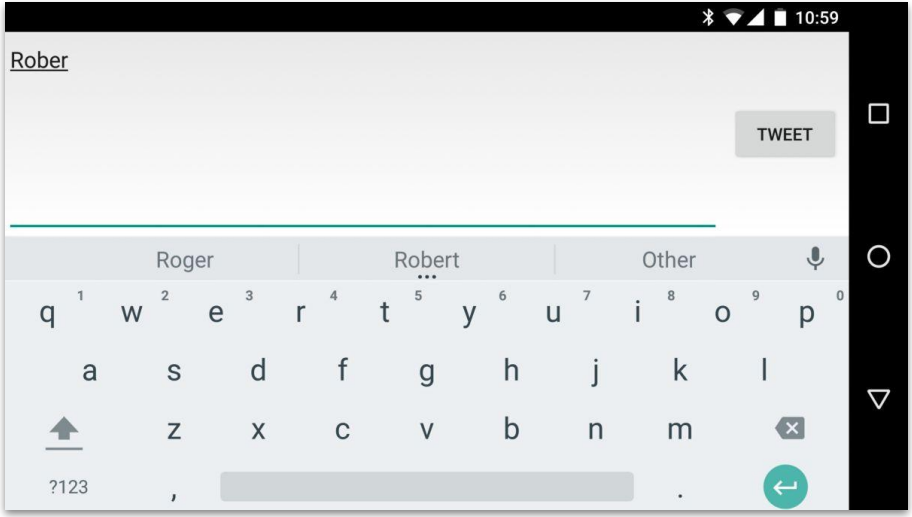

Mobile phone keyboard

Predict next word(s) as we type in...

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Search suggestions

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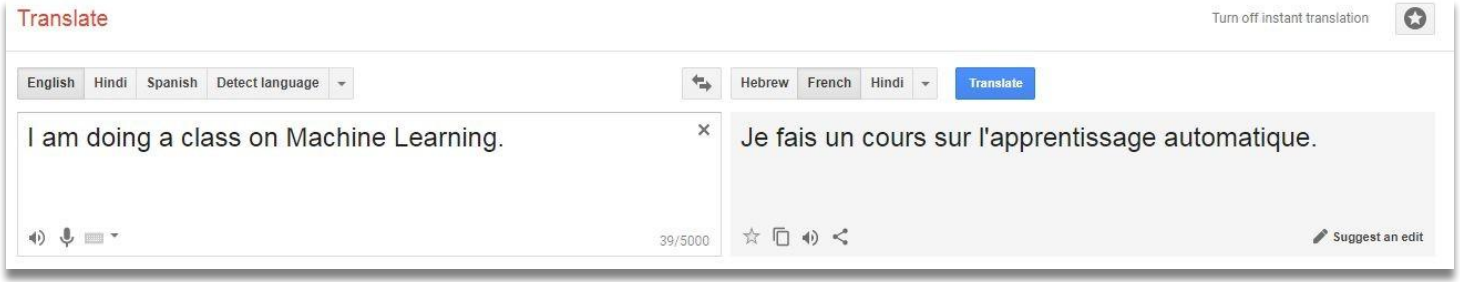

Language translation

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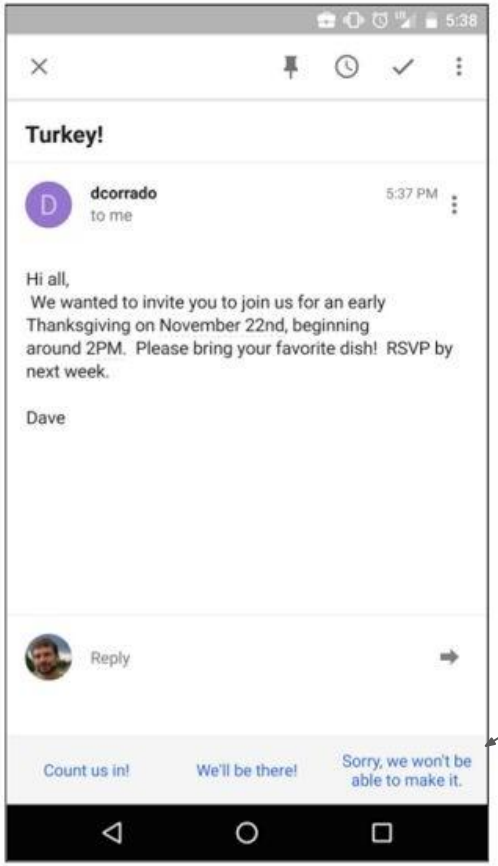
Email Response



Suggested Answers

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Chatbots

Make support available in 24x7 format

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Sentiment analysis

Do people like our products or Services?

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Building NLP Models

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	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1.0	296.0	15.3	396.90	4.98
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2.0	242.0	17.8	396.90	9.14
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2.0	242.0	17.8	392.83	4.03
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3.0	222.0	18.7	394.63	2.94
4	0.06905	0.0	2.18	0.0	0.458	7.147	54.2	6.0622	3.0	222.0	18.7	396.90	5.33
5	0.02985	0.0	2.18	0.0	0.458	6.430	58.7	6.0622	3.0	222.0	18.7	394.12	5.21
6	0.08829	12.5	7.87	0.0	0.524	6.012	66.6	5.5605	5.0	311.0	15.2	395.60	12.43
7	0.14455	12.5	7.87	0.0	0.524	6.172	96.1	5.9505	5.0	311.0	15.2	396.90	19.15
8	0.21124	12.5	7.87	0.0	0.524	5.631	100.0	6.0821	5.0	311.0	15.2	386.63	29.93
9	0.17004	12.5	7.87	0.0	0.524	6.004	85.9	6.5921	5.0	311.0	15.2	386.71	17.10

Boston Housing Price dataset

Structured Data

1. has features (columns)
2. All records have same features
3. Features maintain order across examples

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Hunger Games (2012)

4 of 4 people found the following review helpful

★★★★★ **Recent Convert**, September 18, 2012

By **dharmadude** - See all my reviews

This review is from: Barefoot Running - The Movie: How to Run Light and Free by Getting in Touch with the Earth (NTSC/US Version) (DVD)

As yet, have not viewed film in its entirety but was moved to post a brief note, because thus far, I absolutely LOVE this DVD!! The videography is stunning, the setting on Maui is gorgeous (of course) and the material is very well presented. As a relative new-comer to this barefoot running "thing", I was hoping for some solid, fundamental instruction, as well as inspiration to continue on my fitness path.
I was not disappointed. Having tired of the ever-present aches, sprains & other maladies associated with "normal" distance running, my ethusiasm for running has only recently returned, thanks to the barefoot approach. (sometimes "cheat" with miniamlist shoes) As a result of watching a good portion of this eloquently produced film, I am now fully convinced that I will be a barefoot runner for the duration. Was also quite impressed by the authors, who are a husband & wife team, I think. They exude a truly genuine quality & are clearly passionate about the work they are doing. Not to mention that they appear to be in excellent condition. Guess they practice what they preach.
Now excuse me while I get back to watching the video.

★☆☆☆☆

By Ryan Galaska on February 16, 2016

Verified Purchase

Hunger isn't a game.

0 of 1 people found this review helpful

Textual Data

1. Do not have features like in tabular data.
2. Examples usually have different size.

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What could be the features of textual Data?

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4 of 4 people found the following review helpful

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Now excuse me while I get back to watching the video.

Features in textual Data

1. Words?
2. Characters?
3. Combination of words (n-grams)?
4. What else?

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

Tokenization

Breaking text into tokens e.g words, chars etc

How do we find features of text data?

1. Breakdown text into individual tokens (e.g words)
2. Find the unique tokens
3. Unique tokens form the vocabulary of our dataset

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Document #1

He is a good boy. She is also good.

“He”, “is”, “a”, “good”, “boy”. “She”, “is” “also” “good”

Document #2

Radhika is a good person.

“Radhika” “is” “a” “good” “person”.

Tokenization

Breaking text into tokens e.g words, chars etc

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Document #1

He is a good boy. She is also good.

“He”, “is”, “a”, “good”, “boy”. “She”, “is” “also” “good”

Document #2

Radhika is a good person.

“Radhika” “is” “a” “good” “person”.

Vocabulary

a, also, boy, good, He, is, person, She, Radhika

Build Vocabulary

Unique words in entire dataset

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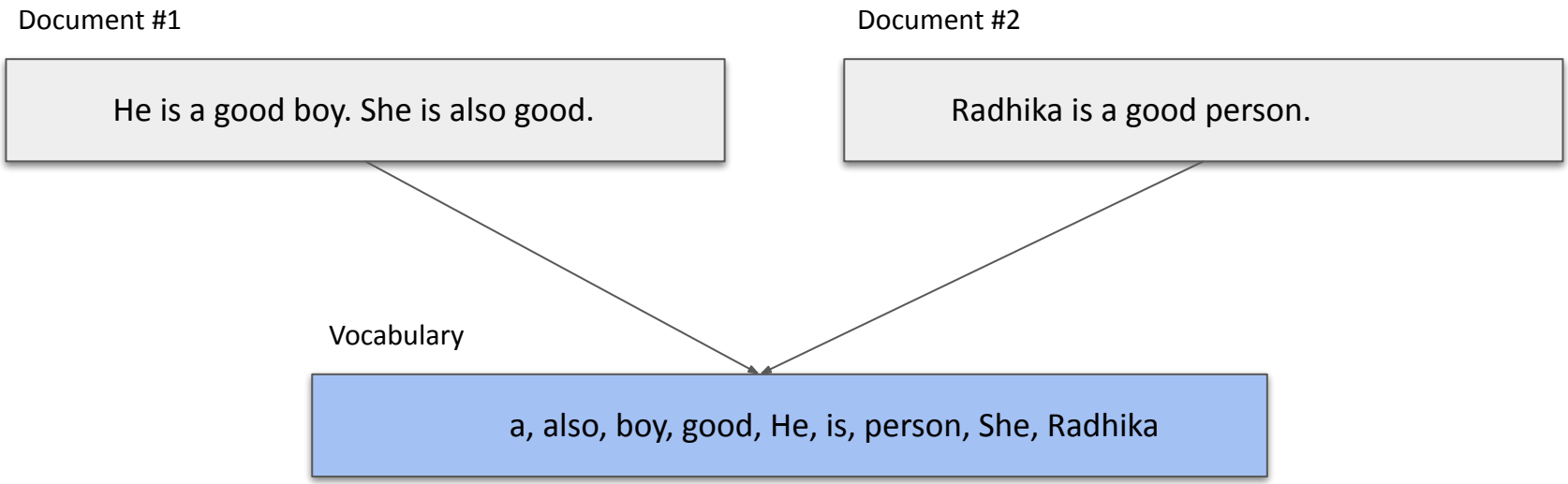

Vocabulary represents the ‘features’ of NLP data

Prepare vocabulary before building an NLP model

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Text Data

"Hyderabad is the capital of Telangana."

→

'Hydeíabad', 'is', 'the', 'capital', 'of', 'Telangana', '.'

Tokenization

Breaking text into tokens e.g words, chars etc

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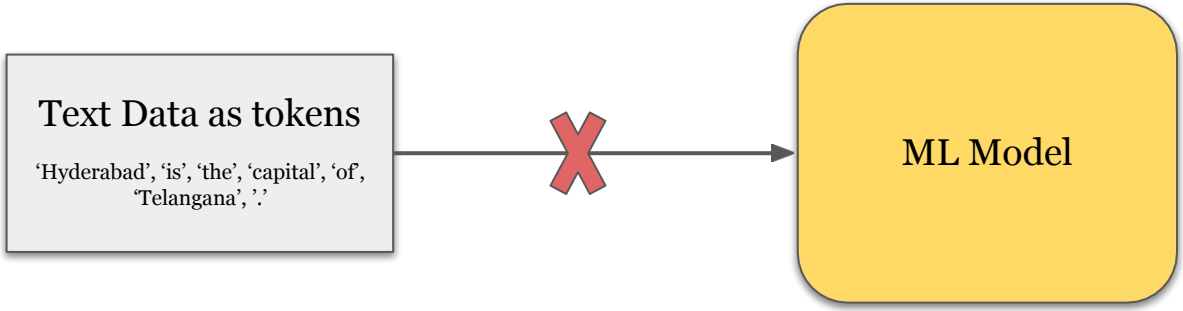

Math works with numbers

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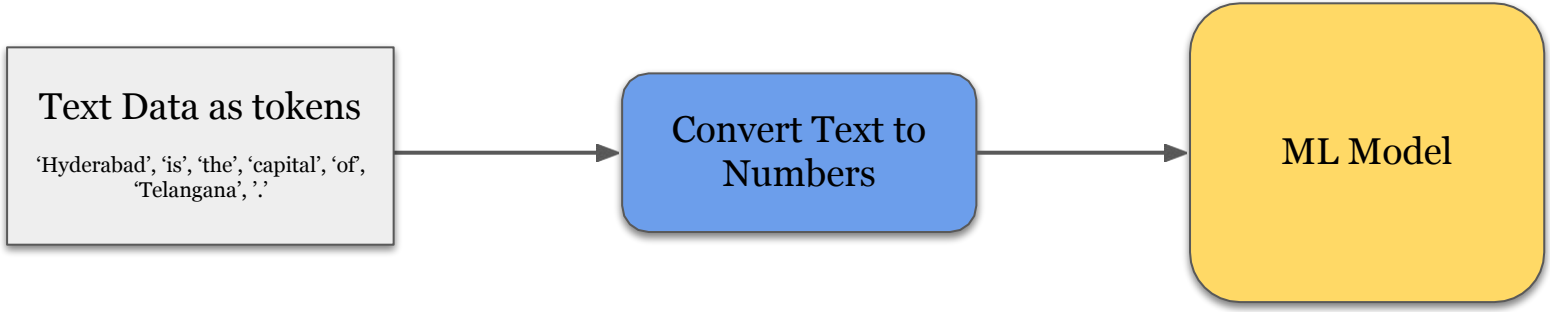

Vectorization

Converting each token into number(s)

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How to convert text into Numbers

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

Vocabulary

a, also, boy, good, He, is, person, She, Radhika

	a	also	boy	good	He	Is	person	She	Radhika
Index	0	1	2	3	4	5	6	7	8

Assign index for each word in vocabulary

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How to convert individual documents into numbers using index and vocabulary?

We have multiple approaches …

1. Bag of words
2. CountVectorizer
3. TF-IDF Vectorizer

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Bag of Words

1. A simple feature extraction approach in NLP
2. Ignores grammar / structure
3. Represents each document by measuring presence of vocabulary words

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

Vocabulary

a, also, boy, good, He, is, person, She, Radhika

	a	also	boy	good	He	is	person	She	Radhika
Index	0	1	2	3	4	5	6	7	8
Document# 1									

Check if each word in Vocabulary appears in Document # 1

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

Vocabulary

a, also, boy, good, He, is, person, She, Radhika

	a	also	boy	good	He	is	person	She	Radhika
Index	0	1	2	3	4	5	6	7	8
Document #1	1	1	1	1	1	1	0	1	0

Indicate '1' if word was found, otherwise indicate '0'

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

Vocabulary

a, also, boy, good, He, is, person, She, Radhika

	a	also	boy	good	He	is	person	She	Radhika
Index	0	1	2	3	4	5	6	7	8
Document #1	1	1	1	1	1	1	0	1	0
Document #2	**1**	**0**	**0**	**1**	**0**	**0**	**1**	**0**	**1**

Our data is now in table format and ready for ML model :)

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Count Vectorizer

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

Vocabulary

a, also, boy, good, He, is, person, She, Radhika

	a	also	boy	good	He	is	person	She	Radhika
Index	0	1	2	3	4	5	6	7	8
Document# 1									

Count how many times each word in Vocabulary appears in Document #1

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

Vocabulary

a, also, boy, good, He, is, person, She, Radhika

	a	also	boy	good	He	is	person	She	Radhika
Index	0	1	2	3	4	5	6	7	8
Document #1	1	1	1	2	1	2	0	1	0

Document # 1 = [1, 1, 1, 2, 1, 2, 0, 1, 0]

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Document #1

He is a good boy. She is also good.

Document #2

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Vocabulary

a, also, boy, good, He, is, person, She, Radhika

	a	also	boy	good	He	is	person	She	Radhika
Index	0	1	2	3	4	5	6	7	8
Document #1	1	1	1	2	1	2	0	1	0
Document #2	1	0	0	1	0	1	1	0	1

Document # 2 = [1, 0, 0, 1, 0, 1, 1, 0, 1]

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

Vocabulary

a, also, boy, good, He, is, person, She, Radhika

	a	also	boy	good	He	is	person	She	Radhika
Index	0	1	2	3	4	5	6	7	8
Document #1	1	1	1	2	1	2	0	1	0
Document #2	1	0	0	1	0	1	1	0	1

Document Term Matrix (DTM)

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TF-IDF Vector

Not just simple counting

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Term Frequency

measures how important a word is for a document

Inverse Document Frequency

How common (or uncommon) a word is across the dataset

TF-IDF

… involves two computations

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Document #1

He is a good boy. She is also good.

He	1
is	2
a	1
good	2
boy	1
she	1
also	1
Total	**9**

$$TF = \frac{Frequency\ of\ the\ word\ in\ a\ Doc}{Total\ number\ of\ words\ in\ the\ Doc}$$

TF(He, doc#1) = 1/9 = 0.11

TF(good, doc#1) = 2/9 = 0.22

TF captures how important a word is to the document (without looking at other documents in the dataset)

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Document #2

Radhika is a good person.

Radhika	1
is	1
a	1
good	1
person	1
Total	**5**

$$TF = \frac{Frequency\ of\ the\ word\ in\ a\ Doc}{Total\ number\ of\ words\ in\ the\ Doc}$$

TF(He, doc#2) = 0/5 = 0

TF(good, doc#2) = 1/5 = 0.2

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Document #1

He is a good boy. She is also good.

He	1
is	2
a	1
good	2
boy	1
she	1
also	1
Total	**9**

Document #2

Radhika is a good person.

Radhika	1
is	1
a	1
good	1
person	1
Total	**5**

$$IDF = log(\frac{Num\ of\ Docs}{Word\ in\ Num\ of\ Docs})$$

IDF(He) = log(2/1) = 0.301

IDF(good) = log(2/2) = 0

IDF tells us if a word (feature) can be used to distinguish documents. If a word appears in majority of the documents then IDF will be close to '0' *i.e.* give low weightage to that feature.

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

He	1
is	2
a	1
good	2
boy	1
she	1
also	1
Total	**9**

Radhika	1
is	1
a	1
good	1
person	1
Total	**5**

$$IDF = log(\frac{Num\ of\ Docs}{Word\ in\ Num\ of\ Docs})$$

IDF(He) = log(2/1) = 0.301

IDF(good) = log(2/2) = 0

TF-IDF(He, doc#1) = 0.11 * 0.301 = 0.03311

TF-IDF(good, doc#1) = 0.22 * 0 = 0

TF-IDF(He, doc#2) = 0 * 0.301 = 0

TF-IDF(good, doc#2) = 0.2 * 0 = 0

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Document #1

He is a good boy. She is also good.

Document #2

Radhika is a good person.

Vocabulary

a, also, boy, good, He, is, person, She, Radhika

	a	also	boy	good	He	Is	person	She	Radhika
Index	0	1	2	3	4	5	6	7	8
Document #1				0	0.03311				
Document #2				0	0				

TF-IDF Vector

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